

Reading the convergence rate off the decomposition structure

treewidth as the easy/hard axis for $Z_L = \sum_{|I|=L} \omega(I)$

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Abstract

Many “rates” are the growth rate $\lambda = \lim_L Z_L^{1/L}$ of a sum over length- L words, $Z_L = \sum_{|I|=L} \omega(I)$. This note makes one claim precise: *the decomposition structure of Z_L – the treewidth of the graph that records which indices its weight couples – tells you the **nature and difficulty** of the rate before you compute a single digit.* It does **not** tell you the numerical value; it tells you whether the rate is an exactly-computable eigenvalue, whether it is likely algebraic or transcendental, what kind of corrections to λ^L to expect (geometric vs. a power law L^θ), and what it costs to get it. The cleanest payoff is at the hard end: when no finite transfer matrix exists, the same picture says how far you can push Z_L and how to *verify a conjectured exponent* – which is exactly how the 2D self-avoiding-walk exponent $11/32$ is corroborated.

1 The question

Write $Z_L = \sum_{I \in \mathcal{W}_L} \omega(I)$ as a tensor contraction (an EINSUM) and let G_L be its *dependency graph*: one vertex per summation index, an edge between two indices whenever they appear together in a factor of ω .¹ The cost of evaluating one Z_L by optimal index elimination is $\Theta(q^{\text{tw}(G_L)+1})$ per site, where q is the alphabet size and tw is the treewidth.² The whole question is how $\text{tw}(G_L)$ behaves as $L \rightarrow \infty$.

Bounded treewidth $\iff$ a finite transfer matrix carries the weight $\iff$ the rate is an eigenvalue.

Growing treewidth $\iff$ no finite carrier $\iff$ the rate is only estimable, and a power law may appear.

The rest of the note (i) proves the one fact that makes the dichotomy sharp, (ii) refines it into three levels, (iii) turns it into a decision procedure you can run on the asymptotics, and (iv) shows the payoff: verifying conjectured orders.

2 The one fact: a finite carrier forbids a power law

Suppose the family is *transfer-decomposable*: there is a fixed nonnegative matrix T and boundary vectors u, v with

$$Z_L = \langle u, T^L v \rangle \quad \text{for all } L. \quad (1)$$

¹This is the decomposition graph of the companion U -statistics paper, here indexed by the growing word length L rather than a fixed kernel order.

²S. A. Markov and Y. Shi, *SIAM J. Comput.* (2008); the treewidth/contraction link is classical in graphical models. Finding the optimal order is NP-hard (Arnborg–Corneil–Proskurowski, 1987), which is why solvers use heuristics.

This is exactly what bounded treewidth buys (for a linearly ordered family): the state is the set of “live” indices at a cut, of which there are at most q^{tw} .

Lemma 1 (finite carrier $\Rightarrow$ pure geometric corrections). *Let (1) hold with T primitive, simple top eigenvalue $\lambda_1 = \rho(T)$, second modulus $|\lambda_2| < \lambda_1$, and u, v not orthogonal to the Perron eigenvectors. With $\delta := |\lambda_2|/\lambda_1 \in [0, 1)$,*

$$Z_L = c \lambda_1^L (1 + O(\delta^L)), \quad \frac{Z_L}{\lambda_1^L} \xrightarrow{L \rightarrow \infty} c > 0.$$

*In particular a finite primitive carrier **cannot** produce a nontrivial power-law factor L^θ .*

Proof. Perron–Frobenius makes λ_1 simple with positive eigenvectors w_R, w_L ($\langle w_L, w_R \rangle = 1$); the spectral decomposition is $T^L = \lambda_1^L \Pi + R^L$ with $\Pi = w_R w_L^\top$ rank one and $\rho(R) = |\lambda_2|$. Then $Z_L = \lambda_1^L \langle u, w_R \rangle \langle w_L, v \rangle + \langle u, R^L v \rangle$, and $\|R^L\| = O(|\lambda_2|^L)$ gives the claim with $c = \langle u, w_R \rangle \langle w_L, v \rangle > 0$. $\square$

The contrapositive is the useful half, because it is something you can *observe*:

Corollary 2 (a power law certifies entanglement). *If Z_L/λ^L does not tend to a positive constant – e.g. it drifts like L^θ with $\theta \neq 0$, or the ratio Z_{L+1}/Z_L approaches its limit only algebraically ($\sim 1/L$) instead of geometrically – then no finite primitive transfer matrix represents the family. The carrier is infinite and its spectrum has no gap (continuous spectrum, or criticality $\delta \rightarrow 1$).³*

Figure 1 shows both sides on elementary, exactly known sequences: the golden-mean count $Z_L = F_{L+2}$ (a *local* constraint – no two adjacent 1s – hence a 2-state carrier) against the central binomial $Z_L = \binom{2L}{L}$ (a *global* constraint – equal numbers of ± 1 – which forces the carrier to track an unbounded height, so there is none finite). The first has $Z_L/\varphi^L \rightarrow \varphi^2/\sqrt{5}$ and a geometric ratio; the second has $Z_L/4^L \sim (\pi L)^{-1/2}$ and a $1/L$ ratio. Same template, opposite classes, and the class is legible in the picture.

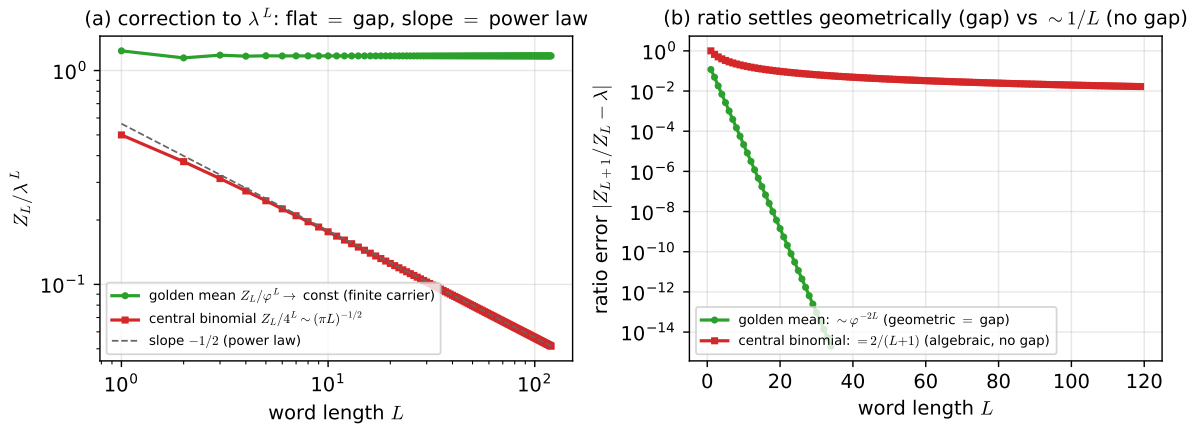


Figure 1: The structural diagnostic (Corollary 2). (a) Z_L/λ^L is *flat* for a finite carrier (golden mean) and a *power law* of slope $-\frac{1}{2}$ for the carrier-less central binomial. (b) the naive ratio Z_{L+1}/Z_L settles *geometrically* when there is a gap and only as $\sim 1/L$ when there is not. Local constraint vs. global constraint = bounded vs. growing carrier = gap vs. no gap.

³If λ_2 sits in a Jordan block the error gains a factor L^{b-1} , but the *leading* term keeps $c > 0$ because primitivity makes λ_1 simple; so a persistent L^θ multiplying λ^L is still impossible.

3 The axis, in three levels

Treewidth is the first-order classifier; the sharp invariant underneath it is “does a transfer operator with a spectral gap exist”. Three levels:

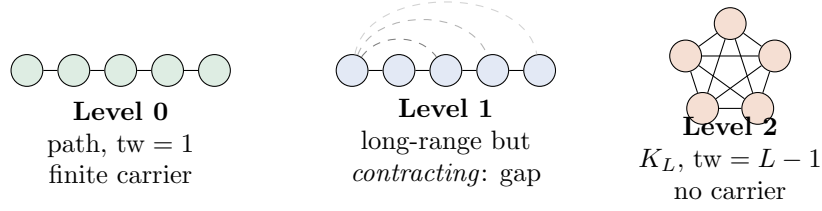


Figure 2: The dependency graph G_L across the three levels: local coupling (path), long-range-but-contracting coupling, and global coupling (complete graph). Treewidth grows left to right.

Level 0 – finite carrier. Bounded treewidth, finite alphabet, weight that couples only nearby indices. The carrier Σ is a *finite* set, $|\Sigma| \leq q^{\text{tw}}$, and (1) holds with a genuine matrix.

- rate $= \rho(T)$, computed once and exactly; **algebraic** if the weights are (a root of $\det(\lambda I - T)$);
- $Z_L = \sum_i c_i \lambda_i^L$: corrections are **geometric**, no power law (Lemma 1);
- cost of Z_L : $O(L q^{\text{tw}+1})$, *linear* in L ; the “divide-out-and-check- $\rightarrow 1$ ” heuristic is a *theorem*.

Examples: subshift entropy, the golden-mean shift, 1D Ising/Potts free energy, and – at each fixed width – a 2D model on a strip.

Level 1 – compact carrier with a gap. The naive treewidth looks unbounded (each weight depends on the whole word through a composition), but the dynamics is *contracting* (hyperbolic / bounded distortion), which collapses the weight onto a transfer *operator* on a function space. That operator is infinite-dimensional but trace-class with a *spectral gap*.

- rate is a “nice” eigenvalue, typically **transcendental**, but computable to arbitrary precision: finite truncations (Chebyshev collocation, Fredholm/spectral determinants) converge **super-exponentially** for real-analytic systems;
- corrections still essentially geometric (the gap);
- this is the regime of *Bowen’s formula* and Ruelle’s thermodynamic formalism.⁴

Examples: $\dim_H E_2 = 0.5312805062772\dots$ (the continued-fraction set),⁵ the Gauss–Kuzmin–Wirsing constant (≈ 0.3036 , literally a spectral gap), conformal Hausdorff dimensions, and top Lyapunov exponents *under a domination/cone condition* – the condition is precisely what restores the gap.

Level 2 – no gap / growing essential treewidth. No transfer operator with a gap exists. Either the treewidth grows ($\text{tw}(G_L) = \Theta(L)$ for global coupling such as K_L , or $\Theta(W)$ on a width- W strip), or the spectrum is continuous / the system is critical.

- the rate is only **estimable**, by brute force over $q^{\Theta(L)}$ words (or $q^{\Theta(W)}$ strip states) followed by series analysis;
- **power-law** (or stranger) corrections appear – the fingerprint of Corollary 2;
- the value can be transcendental, or worse: for the joint spectral radius, deciding $\rho \leq 1$ is *undecidable* and the finiteness conjecture is *false*.⁶

Examples: self-avoiding walks, lattice animals/polyominoes, pattern-avoiding permutations, the affinity dimension of self-affine sets, the joint spectral radius.

⁴R. Bowen (1979); D. Ruelle, *Thermodynamic Formalism*; V. Baladi, *Positive Transfer Operators* (2000).

⁵O. Jenkinson and M. Pollicott, *Adv. Math.* (2001) and the “hundred digits” computation (2018).

⁶V. Blondel and J. Tsitsiklis (2000); T. Bousch and J. Mairesse (2002).

4 So: can you read the rate off the structure?

Its value, no. Its nature, cost, and the right way to attack it, yes. Concretely, the structure fixes four things in advance:

	Level 0	Level 1	Level 2
$\text{tw}(G_L)$ as $L \rightarrow \infty$	bounded	bounded *	growing
rate is. . .	algebraic eigenvalue	transcendental, gapped	estimate only
correction to λ^L	geometric	geometric	power law / stranger
cost of Z_L	$O(L)$	$O(L)$, precision cheap	$q^{\Theta(L)} / q^{\Theta(W)}$
“divide-out” heuristic	theorem	theorem	heuristic (but standard)

* the *combinatorial* treewidth looks unbounded; the *effective* dimension is finite because of contraction.

Diagnostic 3 (a decision procedure). Given Z_L : **(1)** draw G_L and gauge $\sup_L \text{tw}(G_L)$ – local coupling is bounded, global coupling (K_L) grows; **(2)** if the weight is a smooth contraction of an a-priori long word (derivatives, products of matrices through a *linear* functional, conformal maps), suspect Level 1, not Level 2; **(3)** *confirm from the data*: compute Z_{L+1}/Z_L . If it settles *geometrically* you have a gap (Level 0/1) and the rate is an eigenvalue you should compute directly; if it settles only as $\sim 1/L$ (a power law in Z_L) you are in Level 2 and must enumerate and fit.

Step (3) is the bridge between the two halves of Figure 1: the *shape* of the approach to λ is a measurement of the structural class.

5 The payoff: verifying a conjectured order

The most valuable use is exactly where the rate is *not* known: Level 2, where one has a *conjectured* asymptotic

$$Z_L \sim A \lambda^L L^\theta \quad (2)$$

and wants to test it. (In Level 0/1 there is nothing to fit – the rate is an eigenvalue and $\theta = 0$ by Lemma 1.) The standard tool is the *ratio method*: from $r_L = Z_{L+1}/Z_L = \lambda(1 + \theta/L + O(L^{-2}))$,

$$\theta_L := L \left(\frac{r_L}{\lambda} - 1 \right) \xrightarrow{L \rightarrow \infty} \theta, \quad \theta_L^{(1)} := L \theta_L - (L-1) \theta_{L-1} \quad (\text{one Richardson step}) \quad (3)$$

converges to θ faster ($O(L^{-2})$). Figure 3 validates this on two sequences whose exponent is *known*: the central binomial ($\theta = -\frac{1}{2}$) and the Catalan numbers ($\theta = -\frac{3}{2}$); the accelerated estimate pins each to several digits from ~ 100 terms.

The genuinely open / numerically settled problems where this is *the* method:

- **Self-avoiding walks (2D), the exponent $\frac{11}{32}$.** $c_L \sim A \mu^L L^{\gamma-1}$ with $\gamma = \frac{43}{32}$, so $\theta = \gamma - 1 = \frac{11}{32} = 0.34375$, predicted by the Coulomb gas and SLE_{8/3}.⁷ The counts c_L are produced by a *finite-lattice transfer matrix* on a strip of width W – a Level-2 family whose treewidth is the strip width – to $L \approx 80$, and (3) (with differential approximants) corroborates $11/32$.⁸ Here the treewidth *is* the budget: the reachable L , hence the precision on θ , is set by the affordable strip width $q^{\Theta(W)}$.

⁷B. Nienhuis (1982); Lawler–Schramm–Werner (2004). The honeycomb connective constant $\mu = \sqrt{2 + \sqrt{2}}$ is a theorem (Duminil-Copin–Smirnov, 2012); the square-lattice μ and the exponent are not.

⁸I. Jensen (2004); A. J. Guttmann (ed.), *Polygons, Polyominoes and Polycubes* (2009).

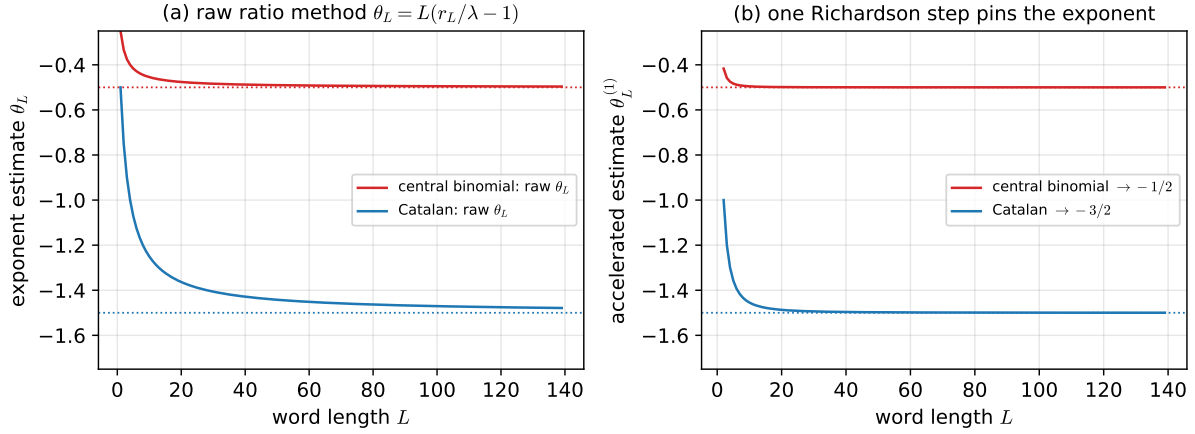


Figure 3: Extracting the order θ from Z_L alone, on cases with a known answer. (a) the raw ratio estimate θ_L ; (b) one Richardson step pins $\theta = -\frac{1}{2}$ (central binomial) and $\theta = -\frac{3}{2}$ (Catalan). The same procedure, run on enumerated counts where λ, θ are unknown, is how conjectured exponents are corroborated.

- **1324-avoiding permutations – a *stretched* exponential.** The Stanley–Wilf limit is $\mu \approx 11.60$, but ~ 50 computed terms reveal the exotic form $B \mu^L \mu_1^{\sqrt{L}} L^g$ – a correction $\mu_1^{\sqrt{L}}$ no finite matrix could ever produce, a vivid instance of Corollary 2.⁹
- **Lattice animals / polyominoes.** Growth constant $\lambda \approx 4.0625697$ with conjectured exponent $\theta = -1$ in 2D, from enumeration to large size and series analysis.¹⁰
- **Joint spectral radius and affinity dimension.** Conjectured extremal values and dimension formulas, tested on finite products T_I ; the honest boundary, where “verify the order” can be provably impossible (undecidability).

The framework’s contribution to this column is threefold: it *certifies* that these problems have no finite carrier (so an exponent $\theta \neq 0$ is expected, not an artifact); it gives the *complexity scale* – the treewidth of the enumeration graph – that fixes how far L , and hence how sharp θ , can go; and the contraction-order optimizers that solve the scheduling half (`opt_einsum`, `cotengra`) are literally the tools that maximize the reachable L .¹¹

6 What the structure does *not* tell you

To keep the claim honest:

- **Not the value.** Two Level-0 families have different eigenvalues; treewidth fixes the *class*, never the number.
- **Combinatorial treewidth can be too pessimistic.** Level 1 is exactly the case where the naive graph looks entangled but a *dynamical* gap (contraction, domination) makes it easy. Detecting that gap is an analytic fact, not a graph computation – it is the rigorous content behind a practitioner’s hope that a weight “approximately factorises”.
- **The fit is heuristic.** In Level 2 the ratio/series method is an extrapolation, not a proof; it can mislead (e.g. a $\sqrt{L}$ stretch hiding as a slowly drifting θ), which is why the 1324 story took many terms.

⁹A. Conway, A. Guttmann and P. Zinn-Justin (2018).

¹⁰I. Jensen (2003).

¹¹D. G. A. Smith and J. Gray (2018); J. Gray and S. Kourtis (2021).

- **Optimal elimination is NP-hard**, and the worst Level-2 problems are genuinely intractable (undecidable joint spectral radius).

In one line: *treewidth* is the axis that says whether a convergence rate collapses to an eigenvalue or must be hunted by enumeration – and in the latter case, how big the hunt is and how to read the conjectured order off the catch.

Reproduce: `python3 code/run_all.py`; Figures 1–3 are `exp5_classification.py` and `exp6_exponent.py`.